

The Distribution of Duration Risk, Fragility, and Central Bank Policy

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Standard asset-pricing logic links financial prices to risk-adjusted expected payoffs.

The 2008–09 crisis challenged this view and motivated *intermediary asset pricing* (Gertler and Kiyotaki, 2010; He and Krishnamurthy, 2013): the balance-sheet conditions of financial intermediaries can generate asset-price dislocations.

Why does this matter?

- **Macroeconomics:** financial-sector conditions affect borrowing costs and, ultimately, real activity (Gilchrist and Zakrajsek, 2012).
- **Asset pricing:** intermediary constraints can explain excess volatility, differences in expected returns, and market-specific frictions arising from segmentation.

Gap: Less is known about how segmentation across financial markets—and monetary policy's effect on that segmentation—transmits to the macroeconomy.

Research question: *How does Treasury-market fragility affect macroeconomic dynamics? What is the role of central banks?*

I focus on the U.S. Treasury market.

[Kashyap et al. \(2025\)](#) model Treasury-market dysfunction as arising from constraints and outside options faced by **banks**, **hedge funds (HFs)**, and **institutional investors (IIs)**, with an important role for derivatives markets and the basis trade.

Institutional investors are major participants in bond/equity markets ($\sim \$21T$), and the Treasury markets ($\sim \$9T$).

- **Why use derivatives?** To manage interest-rate risk with borrowing constrained by regulation ([Alfaro et al., 2024](#); [Jansen et al., 2026](#); [Matt, 2026](#)).
- Their balance sheets are affected by both **conventional** ([Li and Wenning, 2025](#)) and **unconventional** ([Kotronis et al., 2025](#)) monetary policy.

Contribution and outline

Contribution: A general equilibrium model of Treasury intermediation and credit provision:

- Endogenize derivatives demand
- Endogenize the basis trade
- Treasury fragility: from partial eq. to macro GE

Today: Focus on the **partial-equilibrium financial-sector block** and its implications for Treasury-market fragility.

Outline

- ① Intuition, overview, and arbitrage determination
- ② Partial eq. results, does r_t matter?
- ③ Case study: QT and Reserve Management Purchases

Preliminaries: Swaps and the basis trade

Swap contracts is a financial contract in which two parties agree to exchange future cash flows at a predetermined date:

- *Fix leg*: pays predetermined interest rate. *Floating leg*: pays market rate at maturity.
- The pay off of a *fix leg* receiver of a one period swap with notional amount s_t :

$$s_t(r_t^{fix} - r_{t+1}^{float})$$

A (simplified) positive basis is a mispricing in a swap where $r_t^{float} > r_t^{fix}$. How to leverage it?

- At t : borrow today with repayment rate r_{t+1}^{float} and buy a security that pays r_t^{float} , and enter a swap receiving the *floating leg*.
- At $t + 1$, your return is :
$$s_t(r_{t+1}^{float} - r_t^{fix}) + s_t(1 + r_t^{float}) - s_t(1 + r_{t+1}^{float}) = s_t(r_t^{float} - r_t^{fix})$$

Institutional investors and the demand for swaps

Risk averse **institutional investors (IIs)** have a **hedging motive**: longer-duration liabilities than assets.

Example: two assets with payoffs $a : \{a, \kappa a, \kappa^2 a, \dots\}$, $h : \{h, h, h, \dots\}$

IIs	
$a_t q_{a,t}$	$w_{II,t}$
	$h_t q_{h,t}$

$$q_a = \frac{1}{1+r-\kappa}, \quad q_h = \frac{1}{r}$$

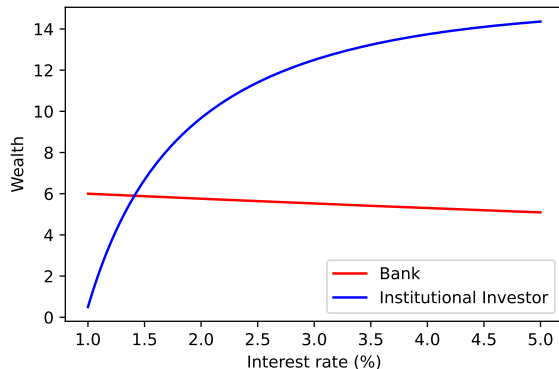
When rates fall:

$$r \downarrow \Rightarrow q_h \uparrow \uparrow > q_a \uparrow > \Rightarrow w_{II} \downarrow \Rightarrow \text{Hedge!}$$

But hedging creates liquidity risk when rates rise: the swap requires cash payments precisely in the high-rate state.

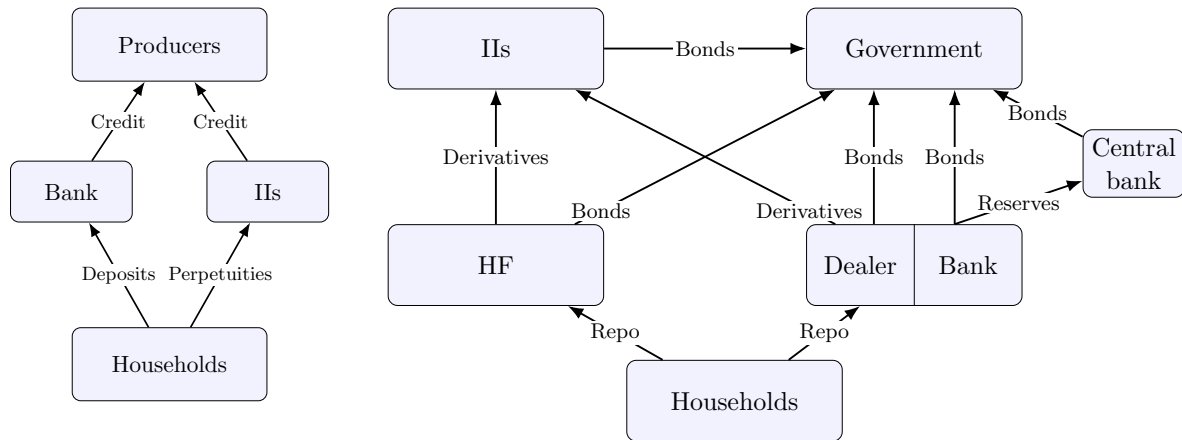
$$m_t = a_{t-1} - h_{t-1} + s(r_{t-1}^{fix} - r_t^{float}) + \Delta a_t q_{a,t} \geq 0$$

► Inst. investor problem



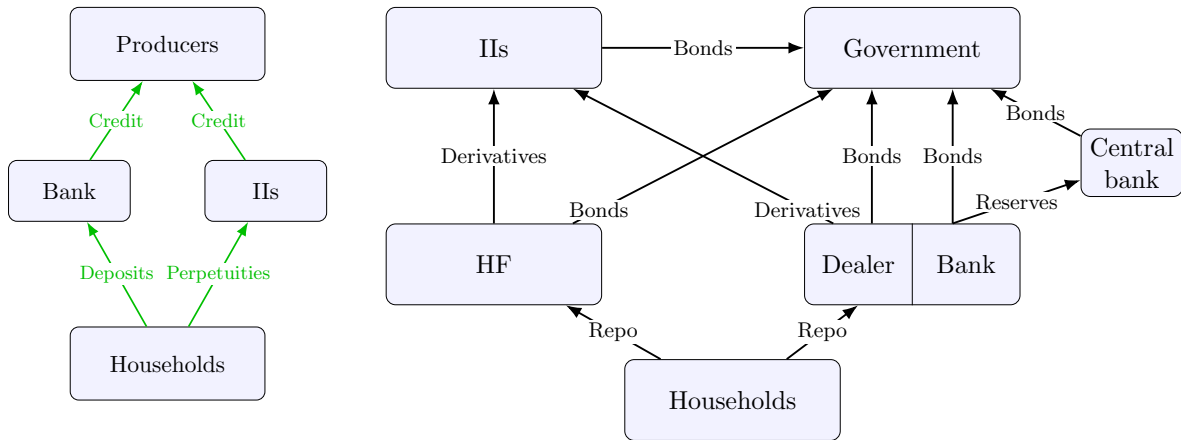
Model overview

A DSGE model linking three financial markets:



Model overview

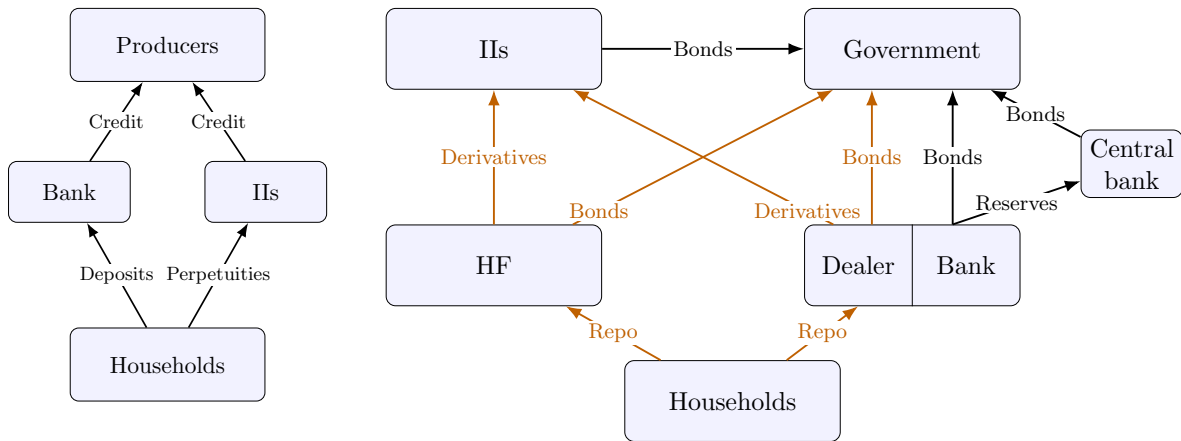
A DSGE model linking three financial markets: Credit supply



(Gertler and Kiyotaki, 2010)

Model overview

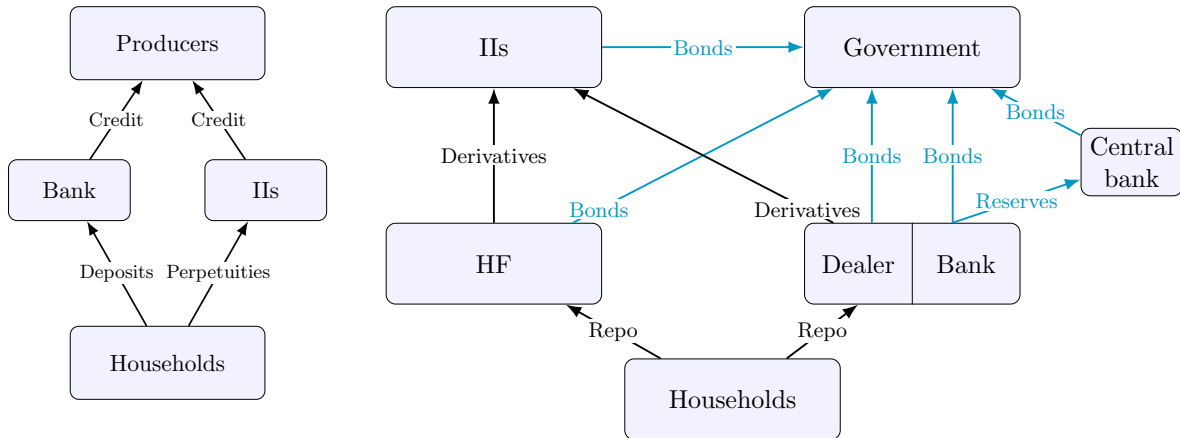
A DSGE model linking three financial markets: **Derivatives mkt**



(Kashyap et al., 2025; Alfaro et al., 2024)

Model overview

A DSGE model linking three financial markets: Treasury mkt



Contribution: GE connection: Credit + Treasury

Banks and basis determination

The outer problem (dynamic)

$$\max_{\{a_{r,t}, b_{b,t}, res_t, x_t, d_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left(m_{0,t+1}^h \left((1 - \phi^b) w_{t+1}^b + \phi^b V_{t+1}^b(w_{t+1}^b) \right) \right)$$

$$q_t a_{r,t} + q_{b,t} b_{b,t} + res_t + x_t = w_t^b + q_t^d d_t$$

$$w_{t+1}^b = \left[R_{t+1}^b q_{b,t} b_{b,t} + R_{t+1}^a q_{t+1} a_{r,t} + (1 + r_t^{IORB}) res_t + V_{t+1}^{deal}(x_t) - d_t \right]$$

$$V_t^b(w_t^b) \geq \eta (q_t a_{r,t} + q_{b,t} b_{b,t} + x_t)$$

Five FOCs, for dealer capacity (x_t):

$$\boxed{\frac{\partial V_{t+1}^{deal}}{\partial x_t} = 1 + r_{t+1}^d + \eta \mu_t}$$

► Preliminaries

The dealer problem (static)

To leverage the basis trade:

① Borrows repo to finance bond purchase:
 $q_{b,t} b_{deal,t} = q_{p,t} p_t^{deal}$

② Hedges in derivatives: $b_{deal,t} = \frac{-s_t^{deal}}{\kappa^b}$

③ Must provide repo collateral (haircut)
 $x_t \geq \gamma q_{b,t} b_{deal,t}$

Using these conditions: ($q_{b,t}^{der} > \mathbb{E}_t q_{b,t}$):

$$V_{t+1}^{deal}(x_t) = x_t \underbrace{\left[\frac{1}{\gamma} \left(\frac{1 + \kappa^b q_{b,t+1}^{der}}{q_{b,t}} - \frac{1}{q_{p,t}} \right) \right]}_{\text{Basis}}$$

Thus:

$$\boxed{\frac{\partial V_{t+1}^{deal}(x_t)}{\partial x_t} = \frac{1}{\gamma} \left(\frac{1 + \kappa^b q_{b,t+1}^{der}}{q_{b,t}} - \frac{1}{q_{p,t}} \right)}$$

Hedge funds: Endogenous absorption capacity

Hedge funds implement the same basis trade as dealers, but can **endogenously raise costly equity** to expand their balance sheet.

$$\begin{aligned} \max_{\{b_{hf,t}, f_t^{hf}, p_t^{hf}, e_t^{hf}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \mathbb{E}_t \left[m_{0,t+1}^h \phi^{hf} w_{t+1}^{hf} \right] \\ \text{s.t.} \quad & w_{t+1}^{hf} = (1 + \kappa^b q_{b,t+1}) b_{hf,t} + f_t^{hf} (q_{b,t+1} - q_{b,t+1}^{der}) - p_t^{hf} - \frac{\psi^{hf}}{2} (e_t^{hf})^2 \\ & q_{b,t} b_{hf,t} = q_{p,t} p_t^{hf} \quad \kappa^b b_{hf,t} = -f_t^{hf} \\ & \gamma q_{b,t} b_{hf,t} \leq (1 - \phi^{hf}) w_t^{hf} + e_t^{hf} \end{aligned}$$

At the leverage constraint:

$$w_{t+1}^{hf} = [(1 - \phi^{hf}) w_t^{hf} + e_t^{hf}] \underbrace{\frac{1}{\gamma} \left[\frac{1 + \kappa^b q_{b,t+1}^{der}}{q_{b,t}} - \frac{1}{q_{p,t}} \right]}_{\text{Basis}} - \frac{\psi^{hf}}{2} (e_t^{hf})^2$$

Hence, the basis determines the incentive to inject equity:

$$(e_t^{hf})^* = \frac{1}{\gamma \psi^{hf}} \left[\frac{1 + \kappa^b q_{b,t+1}^{der}}{q_{b,t}} - \frac{1}{q_{p,t}} \right]$$

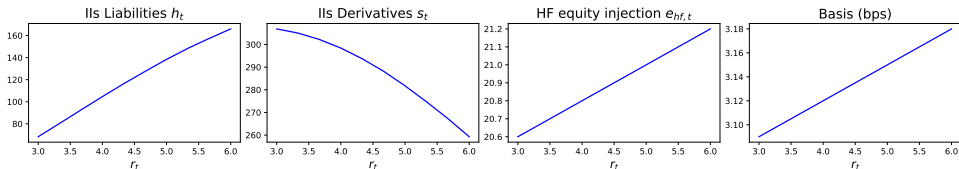
Partial equilibrium dynamics

Setting: at t the CB announces $r_{t+1} \in \{r_t + \epsilon, r_t - \epsilon\}$. All prices are fair.

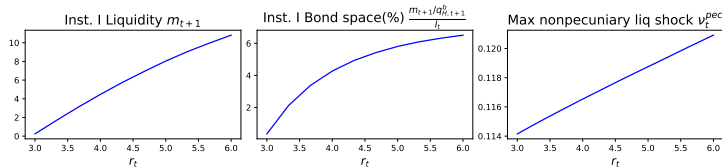
What is the effect of a different starting point r_t ?

Does initial r_t matter?

Greatly. Low r_t enhances solvency risk: large price swings and $\log(\cdot)$ utility $\rightarrow$ hedging motive



What if CB hikes? r_t determines absorption capacity of the system

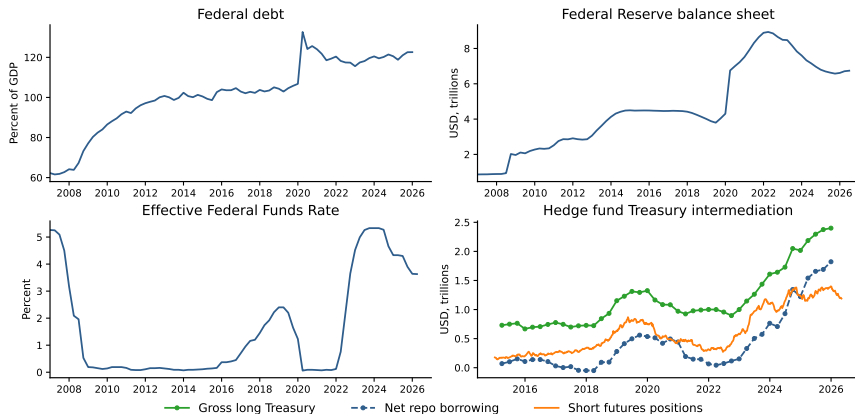


QT space: $m_{t+1} + q_{i,t+1}^b \Delta b_{hf,t}(\nu_t) + \underbrace{q_{i,t+1}^b \Delta b_{b,t}(\nu_t | \mu_t = 0)}_{\text{GE effect}}$

If $\nu_t > \nu_t^{pec}$, (with α_t fixed) bond prices must drop in equilibrium below fair.

Current policy issue

Reserve management purchases (Dec 2025)



Sources: FRED and OFR.

Smaller QT space, in GE: $m_{t+1} + q_{i,t+1}^b \Delta b_{hf,t}(\nu_t) + q_{i,t+1}^b \Delta b_{b,t}(\nu_t | \mu_t = 0) - \Delta b_t^{supply}$
Potential explanations low m_t ? close to $\mu_t > 0$? More sensitivity to ν_t ?

- Treasury absorption capacity depends on who holds what.
- Monetary policy affects this capacity through the balance sheets of **institutional investors, hedge funds, and dealers**.
- This creates a potential trade-off:

Monetary policy $\Rightarrow$ Treasury fragility $\Rightarrow$ credit provision.

Next: Embed this mechanism in general equilibrium and quantify the credit spillovers.

Thank you!

Basic model: Inst. investor problem

$$V_t^d = \max_{\{a_{d,t}, b_{d,t}, s_t, e_t^d\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} m_{0,t+1}^h \frac{(\phi^d w_{t+1}^d)^{1-\sigma^d}}{1-\sigma^d}$$

$$q_t a_{d,t} + q_{b,t} b_{d,t} = (1 - \phi^d) w_t^d + q_{h,t} h_t + e_t^d$$

$$w_{t+1}^d = \frac{1}{\pi_{t+1}} \left[(1 + \kappa^a q_{t+1}) a_{d,t} + (1 + \kappa^b q_{b,t+1}) b_{d,t} + s_t (q_{b,t+1} - q_{b,t+1}^{der}) - (1 + q_{h,t+1}) h_t \right] - \frac{\phi^e}{2} e^{d^2}$$

$$m_t^d = \frac{a_{d,t-1}}{\pi_t} + \frac{b_{d,t-1}}{\pi_t} + \frac{s_{t-1}}{\pi_t} (q_{b,t} - q_{b,t}^{der}) - \frac{h_{t-1}}{\pi_t} + q_{b,t} (\kappa^b \frac{b_{d,t-1}}{\pi_t} - b_{d,t}) + q_t (\kappa^a \frac{a_{d,t-1}}{\pi_t} - a_{d,t}) \geq 0$$

b_t : bond holdings; l_t : long-term liabilities; s_t : swap position; q_t^b, q_t^l : bond and liability prices; m_{t+1} : liquidity position.

Basic model: Hedge fund problem

$$\max_{b_{hf,t}, f_t, e_t} \mathbb{E}_t(w_{t+1}^{hf})$$

$$q_{b,t} b_{hf,t} = w_t^{hf} + e_t - \frac{\phi^{hf}}{2} e_t^2 + q_{p,t} p_t$$

$$\kappa^b b_{hf,t} = -f_t$$

$$\gamma q_{b,t} b_{hf,t} \leq w_t^{hf} + e_t$$

$$w_{t+1}^{hf} = \left(1 + \kappa^b q_{b,t+1}\right) b_t^{hf} + f_t \left(q_{b,t+1} - q_{b,t+1}^{forward}\right) - p_t$$

Repo pricing has liquidity shock $q_t^p = (1 + r_t + \nu_t)^{-1}$, while the forward price is

$q_{b,t+1}^{forward} = E_t q_{b,t+1} + \alpha_t q_{b,t}$, where α_t denotes the basis mispricing.

b_t^{hf} : Treasury holdings; f_t : futures position; q_t^p : repo price; $q_{b,t+1}^{forward}$: forward bond price; γ : leverage requirement; w_t^{hf} : hedge fund wealth; e_t costly equity injection.

Basic model: Bank problem

$$\begin{aligned} \max_{a_t, x_t, d_t} \quad & w_{t+1} = (1 + r_{t+1}^a) a_t + V_{t+1}^{deal}(x_t, \mathbb{D}_t) - (1 + r_{t+1}^d) d_t \\ \text{s.t.} \quad & a_t + x_t = w_t + d_t \\ & w_t \geq \eta(q_{r,t} a_{r,t} + x_t) \end{aligned}$$

► Back

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